

Real-Time Fraud Detection — Business Case

Prepared for Zilch • Head of Fraud • Executive Leadership

Customer Engineering

July 2026

THE PROBLEM ZILCH IS TRYING TO SOLVE

Fraudsters test stolen cards in rapid bursts — often 5–10 low-value transactions within seconds. If your fraud model only sees **stale features**, every transaction in a burst looks clean until it's too late. The loss isn't just the fraudulent spend: it's chargebacks, issuer fees, reputational risk with Thredd, and the engineering cost of building the detection capability from scratch.

WHAT SNOWFLAKE DELIVERS — PROVEN IN TECHNICAL VALIDATION

<2s
 Feature freshness
 out of the box

~14ms
 End-to-end scoring
 per transaction

1
 Platform replacing
 7+ managed services

Zero
 Custom ETL pipelines
 built or maintained

CURRENT COMPLEXITY VS. SNOWFLAKE APPROACH

WHAT ZILCH IS BUILDING

- △ Feature freshness still in development — detection window is undefined until it ships
- △ 7+ services to build, deploy & monitor: Kinesis, Lambda, Glue, RAPIDS, SageMaker, Triton, EKS
- △ Custom preprocessing pipeline needed to align training and serving — drift risk on every retrain
- △ Multiple security boundaries, credentials, and vendor relationships to manage
- △ Engineering bandwidth consumed by infrastructure, not fraud strategy

WHAT YOU GET WITH SNOWFLAKE

- ✓ Sub-2s freshness with no custom build — declare features once, they stream continuously
- ✓ Single platform covers ingestion, feature store, model training, serving, and data warehouse
- ✓ Identical feature logic in training and serving — model drift is structurally impossible
- ✓ One security boundary, one billing relationship, one governance model
- ✓ Fraud team focuses on model quality and thresholds — not infrastructure uptime

PLATFORM SIMPLIFICATION

CURRENT AWS/NVIDIA STACK — 7+ SERVICES

Amazon Kinesis AWS Lambda Amazon S3 AWS Glue ETL
 RAPIDS Preprocessing Amazon SageMaker NVIDIA Triton
 fraud-engine-service Amazon EKS Amazon DynamoDB



SNOWFLAKE — ONE PLATFORM

Snowpipe Streaming Online Feature Store
 Snowpark ML Training Model Registry SPCS Scoring Service

BUSINESS IMPACT

 **Faster Detection = Less Fraud Loss**
 Card-testing bursts are visible from the **second transaction** — within 2 seconds of the first event. A model on stale features cannot see the burst pattern. Every transaction that slips through is a confirmed fraud loss plus chargeback.

 **Engineering Cost Reduction**
 Replacing 10 independently operated services with one platform eliminates cross-service incident response and version management overhead. **Fewer services = fewer pages, less on-call burden.** That headcount redeploy to fraud model iteration.

 **Faster Model Iteration**
 With training data, features, and serving on one platform, the cycle from *new fraud signal identified to model in production* compresses from weeks to days. **Faster iteration directly improves catch rates** as attack patterns evolve.

WHY ACT NOW

Zilch's feature freshness component is still being built. **Every week it remains under construction is a week the fraud model runs blind to intra-session patterns.** Snowflake's Online Feature Store is in public preview today — the technical validation in this document was run on live infrastructure. Zilch can have sub-2-second freshness without waiting for a custom build to ship, and without the ongoing burden of maintaining it.

SUGGESTED NEXT STEPS

1 30-Min Architecture Review
 Map Zilch's Thredd event flow into the Snowflake ingestion path. Confirm data residency and PrivateLink routing.

2 POC on Zilch's Live Data
 Run the validated pipeline on a sample of real transaction history. Measure freshness improvement against current baseline.

3 Commercial Sizing
 Snowflake scales on consumption — no upfront spend. We provide a TCO comparison against the current AWS/NVIDIA stack.